NIFEDIPINE

Avoid immediate-release in hypertension

Tocolytic

CHEMICAL NAME Dimethyl 2,6-dimethyl-4-(2-nitrophenyl)-1,4-dihydropyridine-3,5-dicarboxylate

CLASS Dihydropyridine calcium-channel blocker

TRADE NAMES Adalat · Adalat LA / Retard · Procardia XL · Epilat

THERAPEUTIC USES / INDICATIONS

- Hypertension (modified-release forms).
- Angina (stable and vasospastic).
- Tocolysis in preterm labour (off-label, widely used).
- Severe hypertension in pregnancy (oral, per protocol); Raynaud's phenomenon.

ADMINISTRATION

- **PO:** MR/LA tablets swallowed whole — do not crush or chew.
- Immediate-release capsules: avoid sublingual/bitten use (sudden hypotension, stroke, MI).
- Avoid grapefruit juice.

DOSAGES

- **Hypertension:** MR 20–60 mg once daily (LA 30–90 mg).
- **Angina:** MR 10–40 mg twice daily.
- **Tocolysis (off-label):** oral regimen per local obstetric protocol.
- **Raynaud's phenomenon:** low-dose MR nifedipine (licensed in some countries only).

PHARMACOKINETICS

- **Onset:** IR 20 min; MR 1–2 h. **Duration:** IR 4–8 h; LA 24 h.
- Extensive first-pass metabolism by CYP3A4.
- Excreted in urine. **t½:** 2 h (IR).

PHARMACODYNAMICS

- Blocks **L-type calcium channels** in vascular and uterine smooth muscle → vasodilation, uterine relaxation.
- Reflex tachycardia (especially IR forms).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, flushing.
- Ankle oedema, palpitations.
- Dizziness, nausea.

ADVERSE EFFECTS (SERIOUS)

- Precipitous hypotension with IR forms → myocardial or cerebral ischaemia.
- Reflex tachycardia, worsening angina.
- Gingival hyperplasia.
- Pulmonary oedema (with magnesium sulfate/β-agonists in tocolysis).

CONTRAINDICATIONS & CAUTIONS

- Cardiogenic shock; within 1 month of MI; unstable angina.
- Severe aortic stenosis.
- Use with rifampicin (↓ levels).
- **Caution:** heart failure, hypotension.

KEY INTERACTIONS

- **Magnesium sulfate:** profound hypotension, neuromuscular block.
- **CYP3A4 inhibitors / grapefruit:** ↑ levels.
- **Rifampicin, phenytoin:** ↓ levels.
- **β-blockers:** hypotension, heart failure.

NURSING MANAGEMENT

- Check BP and HR before dose; monitor closely in tocolysis.
- Tocolysis: monitor maternal BP, HR, fetal heart rate and contractions.
- Never crush MR tablets.
- Monitor for oedema, headache.

PATIENT EDUCATION

- Swallow tablets whole.
- Avoid grapefruit juice.
- Report chest pain, fainting, rapid heartbeat.
- Rise slowly.

KEY POINTS

- DHP CCB; use MR forms for hypertension.
- IR forms → dangerous hypotension.
- Also tocolytic in preterm labour.
- Avoid grapefruit.

SOURCES CHECKED **[316]** Nifedipress MR 10 Modified-release tablets (UK SmPC) **[311]** NICE NG133 **[317]** Nifedipine, oral (US label) **[318]** Nifedipine, oral (US label, Alembic Pharmaceuticals Inc)

Full references p. 432–457